The Choice of Normalization Influences Shrinkage in Regularized Regression

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Take-Home Message

Normalization is **not** neutral preprocessing. For **binary features**, the *class balance* (proportion of ones) leaks into your lasso, ridge, and elastic net coefficients—and *which* normalization you pick decides whether it does. **The good news:** the right choice is predictable—match the scaling (δ) to your penalty and goal, or tune it.

The Problem

Regularized models are scale-sensitive, so we *normalize* (center and scale) features first. But there are many ways to do this, and for binary features there is no obvious choice. Despite this, the effect of that choice has gone unstudied.

Below: lasso coefficient paths for a real dataset (w1a) under two normalizations. Standardization makes several features stand out, while max–abs flattens them—the normalization choice alone decides which features look important.

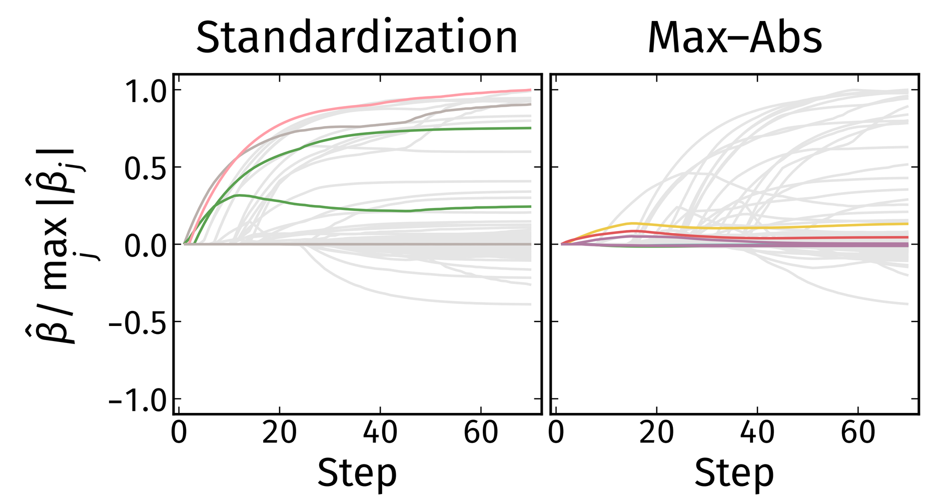

Figure 1: Lasso paths for w1a under standardization ($\delta = \frac{1}{2}$) and max–abs (no scaling, $\delta = 0$), with the first features to enter highlighted. The normalization alone changes which features dominate.

Setup

We study the elastic net (Zou & Hastie, 2005),

$$\frac{1}{2}\|\mathbf{y} - \beta_0 - \tilde{\mathbf{X}}\boldsymbol{\beta}\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1 + \frac{\lambda_2}{2}\|\boldsymbol{\beta}\|_2^2,$$

which gives the **lasso** ($\lambda_2 = 0$) and **ridge** ($\lambda_1 = 0$) as special cases. Each binary feature $\mathbf{x}_j$ is centered and

Class Balance Enters the Estimate

With orthogonal features and no noise, the estimate on the original scale is

$$\hat{\beta}_j = \frac{\operatorname{ST}_{\lambda_1}\left(\frac{\beta_j^* n(q_j - q_j^2)}{s_j}\right)}{s_j\left(\frac{n(q_j - q_j^2)}{s_j^2} + \lambda_2\right)}.$$

The feature variance $q_j - q_j^2$ couples class balance to the estimate, mediated entirely by the scale s_j. The right δ makes it cancel:

Penalty	Cancelling Scale	δ
Lasso	variance	1
Ridge	standard dev.	$\frac{1}{2}$
Elastic net	**none exists**	

For the elastic net, *no* s_j removes the bias—instead weight the *penalty* ($u_j = v_j = (q_j - q_j^2)^\omega$, $\omega = 1$).

Selection Follows δ

Lower δ means imbalanced features are far less likely to be selected; as $q \to 1$, selection probability $\to 0$ for $\delta < \frac{1}{2}$ and $\to 1$ for $\delta > \frac{1}{2}$.

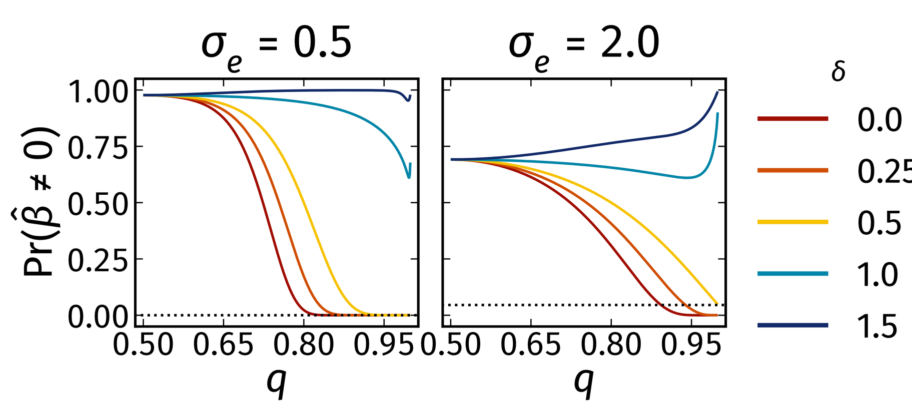

Figure 2: Lasso selection probability vs. class balance q for several δ. Dotted line: asymptotic limit at $\delta = \frac{1}{2}$.

Beyond Orthogonality

The class-balance bias is not just an artifact of the orthogonal theory—it persists with many noisy, correlated features. With 20 signals whose class balance rises 0.5 $\to$ 0.99, no scaling ($\delta = 0$) shrinks the most imbalanced signals to zero, while variance scaling ($\delta = 1$) recovers them.

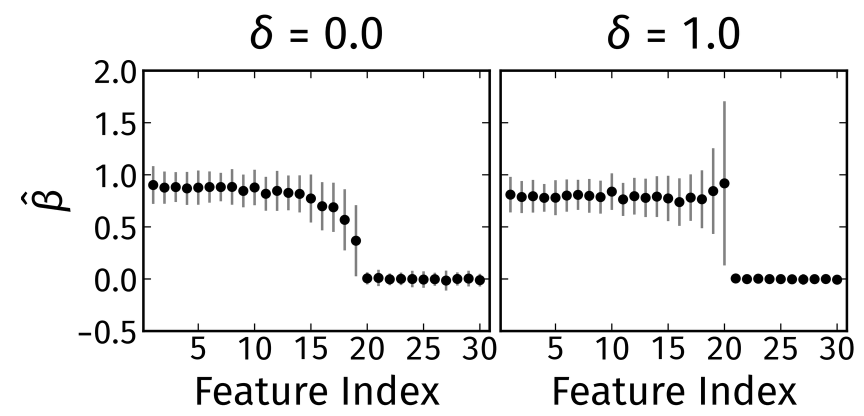

Figure 3: Lasso estimates (mean ± SD) for a simulation with $p = 1000$ (the first 20 features are signals with class balance 0.5 $\to$ 0.99; first 30 shown). No scaling drops the imbalanced signals; variance scaling recovers them.

Tuning δ

Larger δ removes class-balance bias but inflates variance, so there is no universal best δ. On real data, the cross-validated error is minimized at an *interior* δ that differs by dataset—so treat δ as a hyperparameter and tune it.

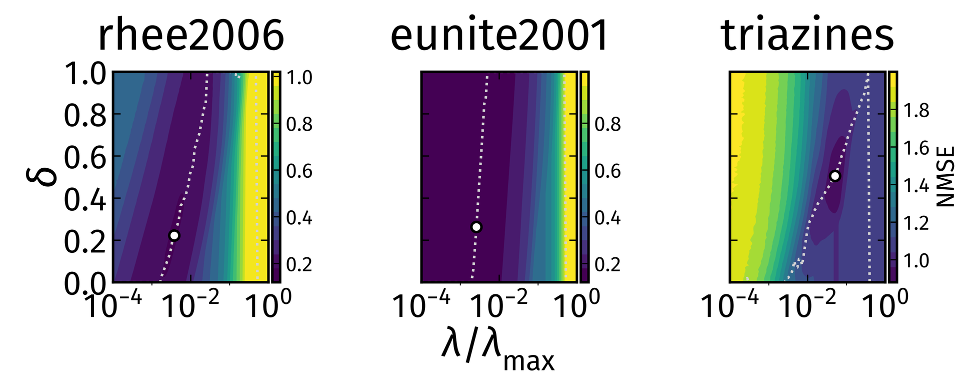

Figure 4: Hold-out (validation) NMSE over (λ, δ) for the lasso on three datasets. White dot: tuned optimum; dotted line: best δ at each λ. The optimal δ is interior and dataset-dependent.

Recommendations

Selection If missing a feature is costly, scale by *variance* ($\delta = 1$).

Prediction Standard deviation or no scaling; better, *tune* δ as a hyperparameter.

Elastic net Weight the *penalty* ($\omega = 1$), not the features.

Why It Matters

The two most common defaults are *both* biased: standardization (the glmnet default; Friedman et al., 2010) and no scaling (recommended for sparse data in scikit-learn). The class-balance effect is therefore likely pervasive in published analyses.

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